Constructing and Certifying Second-Order Euler Distribution Accuracy

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***Abstract*—Explicit Euler can approximate a moving distribution to second order while its trajectories remain first-order accurate, provided the material acceleration preserves each instantaneous marginal. We develop constructive and quantitative criteria for this mechanism on a prescribed probability path. A reference-coordinate equation produces smooth, globally regular nonradial and nonlinear examples that are not affine images of radial or independent-product laws; a variant starts from a standard Gaussian. In a concrete correction family, a quartic minimization recovers the unique zero of an exact residual upper certificate. The weighted-gradient component of acceleration is exactly twice the leading local Wasserstein error coefficient. We use it to prove a finite-time mixed-order bound without requiring a regular flow for the projected solenoidal component, and combine this bound with target-path and velocity-regression errors. Convex-observable and covariance tests provide stronger infeasibility certificates; an explicit smooth path passes all second-moment tests yet has a positive universal residual floor. Reproducible experiments confirm correction recovery, second-order distribution bounds, first-order particle error, optimization failures, residual sensitivity, and a fitted-field error floor. The construction covers a structured class rather than arbitrary learned probability paths, and it establishes no wall-clock advantage over higher-order solvers.**

***Index Terms*—Flow matching, numerical integration, Wasserstein distance, probability transport, error certification.**

Impact statement. Sampling systems are judged by output distributions, which can have different numerical errors from individual trajectories. This work supplies constructions, failure tests, and explicit residual certificates for that distinction. These tools can prevent unsupported claims that a small training loss or divergence penalty guarantees accurate sampling. The present experiments are controlled low-dimensional studies; they do not establish faster image generation or improvements in deployment cost.

I. Introduction

Flow matching and probability-flow models generate samples by integrating a velocity field [1–3]. The numerical objective is often a distribution, even though a solver moves individual particles. Changing how particles are paired across time can therefore change numerical distribution error while preserving the exact marginal at every time. The relevant freedom is the established gauge condition

$$\partial_t p_t + \nabla \cdot (p_t v_t) = 0, \qquad u_t = v_t + r_t, \qquad \nabla \cdot (p_t r_t) = 0.$$

It preserves the entire probability path, not just the endpoints [4–6].

For explicit Euler, the leading omitted trajectory displacement is one half the squared step size times the material acceleration $\mathbf{a}_t = \partial_t u_t + (D_x u_t) u_t$. If this acceleration also preserves the instantaneous distribution, its leading displacement can be removed by changing the coupling between numerical and exact outputs. Under global regularity and moment conditions this gives a finite-time $O(H^2)$ bound in full W_2, where H is the largest step. Particle error can remain $\Theta(H)$.

A conditional error theorem leaves a central question unanswered: which paths admit such a field, and how can an approximate correction be certified? Gauge preservation is linear in a correction, but acceleration preservation is nonlinear. Some paths cannot satisfy it at all. In particular, an evolving path with constant second moment is excluded, and a qualifying gradient field must have straight constant-velocity trajectories, which Euler already integrates exactly. Merely parameterizing divergence-free corrections or plotting the established Gaussian rotation does not resolve this problem.

This paper supplies three complementary answers. First, a linear equation in reference coordinates constructs a family of nonradial, non-affine probability paths with exactly admissible corrections. An explicit two-harmonic example excludes every affine-radial or affine-product representation. Its correction is recovered by a finite global polynomial search; no exact correction is passed to the optimizer. A related construction starts from Gaussian noise. Second, the canonical projected acceleration residual gives a sharp local error coefficient and an operational global bound. Upper and lower residual certificates separate finite-dimensional approximation error from genuine small residual, and the global analysis includes target-path error. Third, convex observables produce path-level obstructions stronger than covariance tests, including an explicit positive residual floor that no correction family can remove.

The scope remains deliberate. We construct a meaningful class of paths rather than solve a general inverse problem for any supplied density. The correction search is finite-dimensional. The empirical learned-field test is a restricted noisy regression model, not a trained image generator. Corrected Euler need not beat Heun in distribution, and it can be substantially worse for common-start particle error. These limitations are part of the results.

II. Setting and the finite-time distribution bound

Let $p_t \in \mathcal{P}_2(\mathbb{R}^d)$ be generated by the exact flow of u. For a grid $0 = t_0 < \cdots < t_N = T$, set $h_k = t_{k+1} - t_k$ and $H = \max_k h_k$. Euler recursively advances $\widehat{X}_{k+1} = \widehat{X}_k + h_k u(t_k, \widehat{X}_k)$ with $\widehat{X}_0 \sim p_0$; write $\widehat{p}_k = \mathrm{Law}(\widehat{X}_k)$. Distribution error uses

$$W_2(\mu,\nu) = \inf_{\gamma \in \Gamma(\mu,\nu)} \left(\int \|x-y\|^2 \, \mathrm{d}\gamma(x,y) \right)^{1/2},$$

whereas common-start particle error pairs numerical and exact trajectories initialized by the same random point. The latter is an upper bound on W_2 and can have a different convergence order.

A field w preserves p in the weighted weak sense if $\int \nabla\phi \cdot w \, \mathrm{d}p = 0$ for every $\phi \in C_c^\infty(\mathbb{R}^d)$. For a positive density this is $\nabla \cdot (pw) = 0$. Define $\mathbf{a} = \partial_t u + (Du)u$ and $\mathbf{j} = \partial_t \mathbf{a} + (D\mathbf{a})u$. The *acceleration condition* is zero weighted divergence of $\mathbf{a}_t$ under every p_t.

Theorem 1 (Finite-time Euler distribution accuracy)**.** *Let p_0 have finite second moment. Suppose u is C^2 on a neighborhood of $[0,T] \times \mathbb{R}^d$, globally L-Lipschitz in space, and $\sup_t \|u(t,0)\| < \infty$. Suppose each $\mathbf{a}_t$ is globally L_a-Lipschitz, satisfies the acceleration condition, and*

$$C_a = \sup_t \|\mathbf{a}_t\|_{L^2(p_t)} < \infty, \qquad C_j = \sup_t \|\mathbf{j}_t\|_{L^2(p_t)} < \infty.$$

Then on every finite increasing grid,

$$\max_k W_2(\widehat{p}_k, p_{t_k}) \le e^{LT} C_* \sum_j h_j^3 \le e^{LT} C_* T H^2,$$
$$C_* = \frac{C_j}{6} + \frac{LC_a}{2} + \frac{TL_aC_a}{8}. \tag{1}$$

The proof uses the frozen acceleration flow Ψ_t^τ. Since it preserves p_t, Euler applied to $\Psi_t^{h^2/2} X$ has the same output distribution as Euler applied to $X \sim p_t$. The displacement $h^2 \mathbf{a}_t(X)/2$ supplied by this shift cancels the leading trajectory defect. Taylor's theorem gives a local $O(h^3)$ coupling error; Euler's Lipschitz factor $1 + Lh$ then gives the displayed bound. Appendix L contains the complete proof and constants. This is the original finite-time result, retained here; the following sections address construction, feasibility, and certification.

The original second-moment identity is

$$m'' = 0, \qquad M'' = 2\mathbb{E}\|u_t\|^2 \ge 2K_{\min}(t), \tag{2}$$

where $m = \mathbb{E}X_t$, $M = \mathbb{E}\|X_t\|^2$, and $K_{\min}$ is the least instantaneous kinetic energy among path generators. Consequently the action of an admissible field is $[M'(T) - M'(0)]/2$. For a minimum-energy gradient generator v, orthogonality gives $\|r_t\|^2_{L^2(p_t)} = M''(t)/2 - \|v_t\|^2_{L^2(p_t)}$ whenever $u = v + r$ is admissible. This budget is an existing consequence, not a new construction. Gradient velocities satisfying all theorem hypotheses have zero acceleration and Euler-exact trajectories (Proposition S10).

III. Feasibility certificates beyond the second moment

The scalar second-moment test is one member of a larger family of path-level restrictions. These restrictions can reject a path before a nonlinear correction search and give positive lower bounds on the best possible acceleration residual. The underlying second-derivative chain rule is standard transport calculus; here it yields certificates that strictly strengthen the second-moment restriction.

For fixed t, define the dual norm

$$\varepsilon_t^{\mathrm{acc}} = \sup_{\eta \in C_c^\infty,\, \|\nabla\eta\|_{L^2(p_t)} > 0} \frac{\left|\int \nabla\eta \cdot \mathbf{a}_t \, \mathrm{d}p_t\right|}{\|\nabla\eta\|_{L^2(p_t)}}. \tag{3}$$

It equals the norm of the weighted-gradient projection of $\mathbf{a}_t$ and is no larger than $\|\mathbf{a}_t - w\|_{L^2(p_t)}$ for every w of zero weighted divergence. This identity alone asserts no regularity of that projection.

Theorem 2 (Convex-observable and weighted-energy certificates)**.** *Assume the regularity and moment hypotheses of Theorem 1, except for the acceleration condition. For $\phi \in C^2(\mathbb{R}^d)$ with bounded Hessian, set $F_\phi(t) = \int \phi \, \mathrm{d}p_t$. Then*

$$F_\phi''(t) = \int u_t^\top D^2\phi \, u_t \, \mathrm{d}p_t + \int \nabla\phi \cdot \mathbf{a}_t \, \mathrm{d}p_t. \tag{4}$$

If the acceleration condition holds and ϕ is convex, F_ϕ is convex in time. More generally, let $K_\phi(t)$ be the infimum of $\int v^\top D^2\phi \, v \, \mathrm{d}p_t$ over finite-energy instantaneous velocities satisfying $\partial_t p_t + \nabla\cdot(p_t v) = 0$. For convex ϕ and $\|\nabla\phi\|_{L^2(p_t)} > 0$,

$$\varepsilon_t^{\mathrm{acc}} \ge \frac{[K_\phi(t) - F_\phi''(t)]_+}{\|\nabla\phi\|_{L^2(p_t)}} \ge \frac{[-F_\phi''(t)]_+}{\|\nabla\phi\|_{L^2(p_t)}}. \tag{5}$$

Unbounded Hessians are permitted when the chain rule is dominated and $\nabla\phi$ belongs to the weighted-gradient closure, including polynomial tests under the corresponding finite moments.

Thus a feasible path satisfies $p_{(1-\lambda)s+\lambda t} \preceq_{\mathrm{cx}} (1-\lambda)p_s + \lambda p_t$: each convex observable at an intermediate time is bounded by its endpoint mixture average. This is necessary, not sufficient. Every straight stochastic interpolation $\mathrm{Law}((1-t)X_0 + tX_1)$ already passes this test. Under weighted Poincare and uniform ellipticity assumptions, K_ϕ is obtained by a linear weighted elliptic problem. Appendix C states these hypotheses and gives the characterization and all proofs.

Proposition 3 (Covariance constraints)**.** *Under the acceleration condition and the preceding hypotheses, let $m(t) = m(0) + t\xi$ and $C(t) = \mathrm{Cov}(X_t) > 0$. There exists B_t such that*

$$C' = B_t + B_t^\top, \qquad \begin{pmatrix} C & B_t^\top \\ B_t & C''/2 \end{pmatrix} \succeq 0. \tag{6}$$

In particular $C'' \succeq 0$, and every directional standard deviation $\sqrt{e^\top C(t) e}$ is convex.

The block-matrix condition is a semidefinite necessary test using only the prescribed covariance and its first two derivatives. It can fail even when the scalar second-moment test passes. The following example passes both tests but is excluded by a fourth-order convex observable.

Proposition 4 (A residual floor beyond covariance tests)**.** *Let $Z \sim \mathcal{N}(0, I_2)$ and $0 \le t \le 1/2$. Define*

$$\kappa(r) = \frac{-1 + 5e^{-r^2/2}}{4}, \qquad b(z) = \kappa(\|z\|)z,$$
$$G_t(z) = z + \frac{t^2}{2} b(z), \qquad p_t = (G_t)_{\#}\,\mathrm{Law}(Z). \tag{7}$$

This path has smooth positive densities and a smooth globally Lipschitz minimum-action generator $v_t(G_t z) = tb(z)$. It passes the scalar second-moment and covariance tests throughout the interval. Nevertheless, no generator with the stated regularity

can satisfy the acceleration condition. Every such generator obeys

$$\varepsilon_0^{\mathrm{acc}} \geq \frac{\sqrt{3}}{16}. \tag{8}$$

The separation is explicit: $M'' - 2K_{\min} = 1/4 + 23t^2/144 > 0$, and $C(t) = (1 + t^2/16 + 23t^4/1152)I_2$ satisfies (6). However, $(\mathbb{E}\|X_t\|^4)'' < 0$ on the entire interval, with value -3 at zero. Since $\|\nabla\|x\|^4\|_{L^2(p_0)} = 16\sqrt{3}$, (5) gives the displayed floor. The obstruction applies to every admissible generator, independently of its parameterization or optimization method. In particular no correction family on this path can achieve $\sup_t \varepsilon_t^{\mathrm{acc}} = O(H)$ as $H \downarrow 0$. By Corollary 10, the same obstruction yields an actual local error floor for every such generator, with $E^u_{0,h}(x) = x + hu(0, x)$:

$$\liminf_{h \downarrow 0} h^{-2} W_2((E^u_{0,h})_{\#} p_0, p_h) \geq \frac{\sqrt{3}}{32}.$$

This local conclusion does not exclude accidental cancellation of terminal integration error.

IV. Constructing a nonradial probability path and its correction

The acceleration condition can be imposed before solving a nonlinear Eulerian equation. Let ρ be a reference law, b a field preserving ρ, and F_t smooth diffeomorphisms. If

$$\begin{aligned} \partial_{tt} F_t(z) &= DF_t(z) b(z), \\ p_t &= (F_t)_{\#}\rho, \\ u_t(F_t z) &= \partial_t F_t(z). \end{aligned} \tag{9}$$

then $\mathbf{a}_t(F_t z) = DF_t(z)b(z)$. For every compactly supported smooth test ϕ, the change of variables gives $\int \nabla\phi \cdot \mathbf{a}_t \, \mathrm{d}p_t = \int \nabla(\phi \circ F_t) \cdot b \, \mathrm{d}\rho = 0$. Thus the linear equation for F_t in Eq. (9) constructs an admissible acceleration. A general nonlinear pushforward does not have this property: the displayed second-derivative identity is essential.

A. An explicit globally regular family

Identify $\mathbb{R}^2$ with $\mathbb{C}$, so $Jz = iz$. Take $\rho = \mathcal{N}(0, I_2)$, $a, T > 0$, and $\epsilon > 0$. Define

$$\begin{aligned} f(z) &= z^2 e^{-|z|^2}, \qquad A_t = e^{(1+i)at}, \qquad B_t = \epsilon e^{(1+i)a\sqrt{2}t}, \\ F_t(z) &= A_t z + B_t f(z), \qquad q = \frac{4\epsilon}{e} e^{a(\sqrt{2}-1)T} < 1. \end{aligned} \tag{10}$$

The source law is ρ; the initial target law is $p_0 = (F_0)_{\#}\rho$, which is already non-Gaussian. Writing $k_t(F_t z) = DF_t(z)Jz$, set

$$v_t(F_t z) = \partial_t F_t(z) - a k_t(F_t z), \qquad u_{\omega,t} = v_t + \omega k_t. \tag{11}$$

Here v is a base generator, not a claimed minimum-action field.

Theorem 5 (A constructive nonradial family)**.** *Under Eq. (10), each F_t is a global smooth bi-Lipschitz diffeomorphism, uniformly on $[0, T]$. The laws $p_t = (F_t)_{\#}\rho$ have smooth positive densities on $\mathbb{R}^2$ and finite moments of every order. For every fixed $\omega \in \mathbb{R}$, u_ω generates exactly this same path, with trajectories*

$$X_t = F_t\big(e^{i(\omega - a)t} Z\big), \qquad Z \sim \rho.$$

The choice $\omega = a$ satisfies the acceleration condition, and all smoothness, growth, global Lipschitz, and moment assumptions of Theorem 1. Consequently its Euler distributions have error $O(H^2)$ on every finite grid. For $\epsilon > 0$, no p_t is an invertible affine image of a radial law or of a product of two independent real laws.

The key identity is

$$\partial_{tt} F_t = 2a^2 \partial_\theta F_t = 2a^2 DF_t Jz.$$

It follows because the angular harmonics z and $f(z)$ have degrees one and two, and their time exponents have squares $2ia^2$ and $4ia^2$. Also $\|Df\|_\infty \leq 4/e$, so $F_t^{-1}(x)$ is the unique limit of the contraction $z_{m+1} = A_t^{-1}(x - B_t f(z_m))$, with contraction factor at most q. Thus the density $p_t(x) = \rho(F_t^{-1}x)/\det DF_t(F_t^{-1}x)$ and the velocity are computable without a PDE solve. The inverse iteration adds real computational cost; the numerical comparison includes it.

This is a family, not just one isolated deformation. Replace $B_t f$ by $\sum_{n=2}^m \epsilon_n e^{(1+i)a\sqrt{n}t} f_n(z)$, where $f_n(e^{i\theta}z) = e^{in\theta} f_n(z)$ and the f_n are smooth rapidly decaying functions. The same construction works whenever

$$\sum_{n=2}^m |\epsilon_n| e^{a(\sqrt{n}-1)T} \|Df_n\|_\infty < 1.$$

Arbitrary smooth radial profiles times z^n give such harmonics. This constructs a specified class of paths. It does not solve the inverse problem for an arbitrary externally supplied density path. Appendix A proves all assertions, including the nonproduct certificate.

Corollary 6 (A Gaussian starting law)**.** *Replace B_t by $\epsilon \sinh((1+i)a\sqrt{2}t)$ and assume $(4/e)\epsilon\sqrt{2}\sinh(a\sqrt{2}T) < 1$. Then F_0 is the identity, so $p_0 = \mathcal{N}(0, I_2)$, and the acceleration-preserving construction and its $O(H^2)$ bound still hold. For every $t > 0$ the target law is neither affine-radial nor affine-product.*

This follows from $B_t'' = 4ia^2 B_t$ and $|\sinh((1+i)s)|^2 = \sinh^2 s + \sin^2 s \leq 2\sinh^2 s$. The numerical study below uses the exponential-coefficient family of Eq. (10); the Gaussian-start variant is an additional proved construction, not a separately reported empirical result.

B. Discovering the correction by a finite global search

For the two-mode family let $F_{1,t} = A_t z$, $F_{2,t} = B_t f(z)$, and

$$\zeta_n = a\sqrt{n} + i\{a\sqrt{n} + n(\omega - a)\}.$$

Then $\mathbf{a}_\omega(t, F_t z) = \sum_{n=1}^2 \zeta_n^2 F_{n,t}(z)$. Every ck_t preserves p_t, so the following is an upper certificate for the projected acceleration residual:

$$\begin{aligned} R_t(\omega, c)^2 &= \|\mathbf{a}_\omega - ck_t\|^2_{L^2(p_t)} \\ &= \sum_{n=1}^{2} M_n(t)|\zeta_n^2 - inc|^2, \\ M_1(t) &= 2e^{2at}, \qquad M_2(t) = \frac{8}{125}\epsilon^2 e^{2a\sqrt{2}t}. \end{aligned} \tag{12}$$

Angular orthogonality makes this formula exact for the continuum law. It avoids numerical differentiation and a weighted Poisson solve.

Proposition 7 (Global correction recovery in the stated family)**.** *At any fixed t, minimizing Eq. (12) over $(\omega, c) \in \mathbb{R}^2$ has the unique zero $(a, 2a^2)$. For a fixed ω the unique minimizing coefficient is*

$$c_*(\omega) = \frac{\sum_{n=1}^2 nM_n \operatorname{Im}(\zeta_n^2)}{\sum_{n=1}^2 n^2 M_n}.$$

After substitution, $R_t(\omega, c_*(\omega))^2$ is a coercive quartic polynomial. Its global minimum is found by evaluating its real cubic stationary points; on a bounded interval include the endpoints. The same statement holds for a time-integrated squared residual after replacing $M_n(t)$ by their positive integrals.

The proof is elementary and useful operationally. A zero in mode one forces $\omega = \pm a$ and $c = 2a\omega$. Mode two excludes $\omega = -a$. The leading quartic coefficient is $\sum n^4 M_n > 0$, while c_* is affine in ω. A local optimizer can converge to a nonzero residual minimum; the finite global search resolves this failure within this one-parameter correction family. Floating-point root enumeration is validated against the exact sum-of-squares identity, not presented as an interval-arithmetic proof.

C. An explicit distribution coupling and a particle limit

Theorem 8 (Phase coupling for the constructed field)**.** *For u_a, a uniform grid $h = T/N$, and Euler initialized at $F_0(Z)$, there is a constant C independent of N such that*

$$W_2(\widehat{p}_N, p_T) \le \|\widehat{X}_N - F_T(e^{-ia^2Th}Z)\|_{L^2(\rho)} \le Ch^2. \tag{13}$$

Meanwhile the common-start particle error has the strictly positive limit

$$\lim_{h\downarrow 0} \frac{\|\widehat{X}_N - F_T(Z)\|_{L^2}}{h} = a^2 T \|DF_T(Z)JZ\|_{L^2(\rho)} > 0. \tag{14}$$

The comparison curve $F_t(e^{-ia^2ht}Z)$ solves the explicit modified field $u_a - (h/2)\mathbf{a}_{u_a}$, which generates the same marginals. A second-order Taylor expansion cancels Euler's quadratic local defect, giving the strong $O(h^2)$ comparison in Eq. (13); no infinite modified-equation series is used. This gives a computable coupling bound for a genuinely nonradial law. It is an upper bound on full W_2, not its exact value. The theorem does not claim a positive h^2 limit for that optimal distance.

V. Constructing and certifying approximate corrections

The useful residual is the component of acceleration that changes the distribution. This component has a canonical definition even when its orthogonal complement does not have a classical flow. For a probability measure p, write

$$\mathcal{G}_p = \overline{\{\nabla\phi : \phi \in C_c^\infty(\mathbb{R}^d)\}}^{L^2(p)}$$

and let Π_p be the orthogonal projection onto this space. Its orthogonal complement consists exactly of the fields with zero weighted divergence. This is the standard tangent space decomposition in Wasserstein geometry [4, Chapter 8]; the contribution below is its quantitative use in the Euler error bound. Define

$$\varepsilon_t = \|\Pi_{p_t}\mathbf{a}_t\|_{L^2(p_t)} = \sup_{\substack{\phi\in C_c^\infty \\ \|\nabla\phi\|_{L^2(p_t)}\le 1}} \left| \int \mathbf{a}_t \cdot \nabla\phi \, \mathrm{d}p_t \right|. \tag{15}$$

Thus $\varepsilon_t = 0$ is equivalent to the acceleration condition. No density or Poincaré inequality is needed for this definition.

Here canonical refers to the projection for the chosen generator u, not an intrinsic acceleration of the probability path. For the expanding circular Gaussian, $u(x) = ax$ gives $\Pi_p\mathbf{a} = a^2x$, whereas $u(x) = a(I+J)x$ gives $\Pi_p\mathbf{a} = 0$ on the same path. Intrinsic Wasserstein covariant acceleration instead first uses the minimum-energy gradient velocity $\Pi_p u$.

A. A residual theorem without a regular projected flow

Theorem 9 (Projected acceleration controls Euler error)**.** *Keep the smoothness, spatial Lipschitz, growth, and moment assumptions of Theorem 1, including the common Lipschitz constant L_a of $\mathbf{a}_t$, but do not impose the acceleration condition. Set*

$$\begin{aligned} E_H &= \exp(L_a H^2/2), \qquad B_H = (1+LH)E_H, \\ D_H &= \frac{C_j}{6} + E_H\left(\frac{LC_a}{2} + \frac{HL_aC_a}{8}\right). \end{aligned} \tag{16}$$

Then on every finite increasing grid,

$$\begin{aligned} \max_k W_2(\widehat{p}_k, p_{t_k}) &\le e^{LT}\left(D_H\sum_j h_j^3 + \frac{B_H}{2}\sum_j h_j^2\varepsilon_{t_j}\right) \\ &\le e^{LT}T\left(D_H H^2 + \frac{B_H}{2}\varepsilon H\right). \end{aligned} \tag{17}$$

The second line assumes $\sup_t \varepsilon_t \le \varepsilon$. In particular, the projected solenoidal field $\mathbf{a}_t - \Pi_{p_t}\mathbf{a}_t$ need not be continuous or Lipschitz.

Proposition S12 required a regular flow for the distribution-preserving part of acceleration. Theorem 9 instead follows the full acceleration field and bounds the change of its distribution by the projected residual. The price is the explicit factor $B_H = 1 + O(H)$ and a different cubic constant; no elliptic regularity of the projection is required. Appendix B proves the result. A certificate $\varepsilon_{t_j} \le \kappa H$ at every Euler node gives an $O(H^2)$ bound when the other constants are uniform. The certificate

may belong to one fixed field or to a family of fields indexed by H; these are different assertions. A small value only at a training grid does not control a different evaluation grid without an interpolation bound.

Corollary 10 (Exact local Wasserstein coefficient)**.** *Under Theorem 9, at every fixed $t < T$,*

$$\lim_{h\downarrow 0} \frac{W_2((E_{t,h})_{\#}p_t, p_{t+h})}{h^2} = \tfrac{1}{2}\|\Pi_{p_t}\mathbf{a}_t\|_{L^2(p_t)}. \tag{18}$$

The projected residual is therefore the exact leading local error coefficient. This strengthens the local necessity result, but does not rule out cancellation between different steps at the final time.

B. Upper and lower certificates for the residual

A finite test space detects only part of the gradient projection. Its norm is a lower bound, so reducing it alone cannot certify a small residual. The following identity supplies both sides using a gradient approximation and an independently admissible solenoidal field.

Proposition 11 (Primal and dual residual certificates)**.** *Fix p and $a \in L^2(p;\mathbb{R}^d)$. For any $q \in \mathcal{G}_p$ and $w \in \mathcal{G}_p^{\perp}$, put*

$$\mathcal{L} = 2\langle a, q\rangle_{L^2(p)} - \|q\|^2_{L^2(p)},$$
$$\mathcal{U} = \|a - w\|^2_{L^2(p)}.$$

Then

$$\max(0, \mathcal{L}) \le \|\Pi_p a\|^2_{L^2(p)} \le \mathcal{U},$$
$$\mathcal{U} - \mathcal{L} = \|a - w - q\|^2_{L^2(p)}. \tag{19}$$

For an exact Galerkin solution $q_N = \sum_i c_i \nabla\phi_i$, where $A_{ij} = \langle\nabla\phi_i, \nabla\phi_j\rangle$ and $b_i = \langle a, \nabla\phi_i\rangle$, $Ac = b$ gives $\mathcal{L} = \|q_N\|^2 = b^{\top}A^{-1}b$ when A is positive definite. The general certificate remains valid for inexact coefficients.

This Hilbert-space identity turns a numerical projection into an upper certificate for Eq. (17). For fixed acceleration, gradient and solenoidal linear bases give two quadratic least-squares problems; the gap checks their resolution. An explicit admissible w already gives an upper bound, so a Poisson solve is optional.

A weighted Poisson formulation, an a posteriori Poincaré estimate, and explicit bounds for quadrature and acceleration approximation errors appear in Appendix B. A raw divergence penalty or Monte Carlo standard error alone is not a deterministic residual certificate.

C. A finite-dimensional construction with explicit scope

In two dimensions let J denote the quarter-turn matrix. For a known positive density, stream potentials give the established gauge parameterization [6]

$$r_\theta(t,x) = p_t(x)^{-1} J\nabla_x\psi_\theta(t,x),$$
$$\psi_\theta = \sum_{\ell=1}^{m}\theta_\ell\psi_\ell, \qquad \nabla\cdot(p_t r_\theta) = 0. \tag{20}$$

Compactly supported potentials avoid dividing by arbitrarily small density in the tails. The proof appendix gives the antisymmetric-tensor version in higher dimensions. A finite family need not represent every admissible field.

For a base field satisfying the theorem's bounds, a positive C^4 density, uniformly compactly supported C^4 potentials and a compact parameter set, the corrected fields preserve the exact path and have uniform regularity bounds. Their objective $\mathcal{E}(\theta) = \sup_t \|\Pi_{p_t}\mathbf{a}_{u_\theta,t}\|$ is Lipschitz and attains a minimum. Proposition S2 in the proof appendix gives the assumptions and a certified finite-net search guarantee: parameter mesh δ and objective certificate width η achieve residual at most $\min_\Theta \mathcal{E} + K_\Theta\delta + \eta$.

This finite-dimensional guarantee does not imply efficiency or convergence of a local nonlinear optimizer. A positive minimum in one family does not prove path infeasibility. The appendix also certifies the continuous time supremum from a time grid.

D. Separating path error from integration error

Let p_t now be a target path generated by v, and let u be the field actually integrated. Its exact path need not equal p_t. Besides the acceleration residual under the target, define the path defect

$$\beta_t = \|\Pi_{p_t}(u_t - v_t)\|_{L^2(p_t)}. \tag{21}$$

This defect ignores velocity differences that preserve the target continuity equation. Both diagnostics in the next theorem are evaluated under the target p_t; no density substitution for an unknown learned path is made.

Theorem 12 (Target-path, residual, and integration errors)**.** *Let p_t be generated by a smooth, uniformly spatially Lipschitz v from $p_0 \in \mathcal{P}_2(\mathbb{R}^d)$, with $\sup_t \|v_t\|_{L^2(p_t)} < \infty$. Let u satisfy the smoothness, growth, and Lipschitz assumptions of Theorem 9. Write $F^u_{t,s}$ for its exact flow and assume*

$$C_a = \sup_t \|\mathbf{a}_{u,t}\|_{L^2(p_t)} < \infty,$$
$$C_j^{\diamond} = \sup_{0\le t\le s\le T} \|\mathbf{j}_u(s, F^u_{t,s}(\cdot))\|_{L^2(p_t)} < \infty.$$

Suppose β_t is integrable. Let $\widehat{q}_k$ be Euler's laws for u from any $q_0 \in \mathcal{P}_2$. With $\varepsilon_t = \|\Pi_{p_t}\mathbf{a}_{u,t}\|$ and with D_H in Eq. (16) using $C_j^{\diamond}$ in place of C_j,

$$\max_k W_2(\widehat{q}_k, p_{t_k}) \le e^{LT}\Big[W_2(q_0,p_0) + D_H\sum_j h_j^3 + \frac{B_H}{2}\sum_j h_j^2\varepsilon_{t_j} + \int_0^T \beta_t\,\mathrm{d}t\Big]. \tag{22}$$

The mixed jerk assumption concerns exact u-trajectories started from the target at any time. It is automatic from a uniform linear-growth bound on $\mathbf{j}_u$ and bounded target second moments, but does not follow from a training loss alone. If u generates p_t, then $C_j^{\diamond} = C_j$ and $\beta_t = 0$, recovering Theorem 9.

For conditional flow matching with $v_t(x) = \mathbb{E}[Y_t \mid X_t = x]$ and $\mathrm{Law}(X_t) = p_t$, the population excess squared regression loss is $\mathcal{R}_{\mathrm{excess}} = \int_0^T \|u_t - v_t\|^2_{L^2(p_t)}\,\mathrm{d}t$. Projection contraction

gives $\int_0^T \beta_t \, \mathrm{d}t \le \sqrt{T\mathcal{R}_{\mathrm{excess}}}$. These population and exact-flow facts are standard [1, 7]; Eq. (22) combines them with the Euler residual. There is no finite-sample learning guarantee. An exactly target-solenoidal correction leaves β_t unchanged, but can change the other constants.

For a stream correction formed with an approximate density, the resulting path defect can be bounded using the score discrepancy and a Poincaré constant; Eq. (27) gives the precise bound.

VI. Experiments: discovery, residuals, and cost

We test the nonradial path in Eq. (10) with $a = 0.4$, deformation amplitude 0.18, and $T = 1$. The global contraction bound is below 0.313, so every field evaluation uses a convergent inverse-map iteration. The path, correction family, and fitted correction stay fixed as the Euler grid is refined. Results are deterministic population quadratures, except for the separately identified noisy regression experiment.

a) Correction discovery and certification.: We eliminate the solenoidal coefficient c from the exact residual formula in Proposition 7, enumerate every real root of the remaining quartic's derivative, and evaluate the original sum of squares at each root. This recovers $(\omega, c) = (0.4, 0.32)$ to floating-point precision without initializing at the solution. The coefficient identity in the proposition certifies the exact zero; the approximately 2.1×10^{-16} numerical residual is a floating-point evaluation, not an interval enclosure. A local least-squares run started at $\omega = -0.3$ instead converges to $\omega \simeq -0.392394$, with residual upper bound 0.07241. Thus a successful local optimization termination does not establish feasibility. Global enumeration takes about 0.01 seconds including three local comparison runs.

b) Distribution and particle errors.: Figure 1 reports the second-moment lower bound and the rotation-coupling upper bound from Theorem 8. We evaluate these continuum bounds by 64-point radial Gauss–Laguerre quadrature and 64 equally spaced angles. They are not exact evaluations of W_2, and the quadrature is not interval certified. At $N = 512$, corrected Euler gives bounds 3.439×10^{-7} and 4.881×10^{-7}, with observed orders 1.9983 and 1.9992 from $N = 256$ to 512. Its common-start particle error is 6.609×10^{-4}, with order 0.9992. Base Euler's lower and upper bounds both have order 0.9991. Refining to 96 radial nodes and 128 angles changes the corrected upper bound by less than 8×10^{-17}. Independent DOP853 trajectories at four intermediate times agree with the analytic flows within 6.5×10^{-12} on the tested labels. These checks support the exact path-preservation identity; they do not replace it.

c) Residuals, infeasibility, and model error.: Changing ω by -2% or $+2\%$ gives exact residual upper certificates 0.01345 and 0.01372 at T, and the finest observed upper-bound orders become 1.036 and 0.961. On the infeasible Gaussian path $\Sigma_t = \mathrm{diag}(1 + t, 1 + t + 2t^2)$, the scalar energy budget is positive throughout $[0, 1]$, whereas the first coordinate gives the all-field lower bound $\varepsilon_t \ge [4(1 + t)^{3/2}]^{-1}$. A separate one-coefficient noisy velocity regression over three seeds has analytically known learned paths. Refining Euler reduces its integration error while target error approaches the nonzero continuous-path error. This is a restricted parametric learning check, not a neural generative-model benchmark.

TABLE I
Nonradial path at equal 384 field evaluations. These are quadrature evaluations of continuum W_2 bounds.

Method	W_2 lower	W_2 upper	Particle RMS
Corrected Euler	6.111×10^{-7}	8.675×10^{-7}	8.810×10^{-4}
Base Heun	6.073×10^{-7}	6.085×10^{-7}	6.085×10^{-7}
Base RK3	1.067×10^{-9}	1.070×10^{-9}	1.070×10^{-9}

d) Cost and unfavorable baselines.: Table I preserves the comparison at 384 field evaluations. Heun's evaluated upper bound is slightly below corrected Euler's evaluated lower bound, providing numerical evidence that Heun is more accurate on this path. RK3 is substantially better. Batch runtimes are comparable, and every stage pays for the inverse map; no sampling speedup is established. The correction improves Euler's distribution order, but it is not the best solver in this experiment. Supplement Section D gives timings, inverse tolerances, refinement data, optimizer failures, model-error floors, and reproduction commands.

VII. Established examples and solver trade-offs

The Gaussian and nonlinear radial positive controls are retained with their complete proofs and numerical receipts in the supplement. For $p_t = \mathcal{N}(0, e^{2at} I_2)$, the base field ax and the corrected field $a(I + J)x$ have identical exact marginals. The correction changes Euler's W_2 error from first to second order and doubles the leading common-start particle-error coefficient. With N uniform Euler steps and $N/2$ base-field Heun steps, both methods use N evaluations and have exactly identical Gaussian output distributions at their shared grid points (Proposition S11). Thus this example does not show an advantage over using a second-order solver.

For the shape-changing radial field $\alpha(r)x$, where $\alpha(r) = a + \eta r^2/(1 + r^2)$, the correction $\beta(r)Jx$ with $\beta^2 = \alpha^2 + \alpha r \alpha'$ cancels radial acceleration. Theorem S7 proves strict distribution-error improvement on every grid and positive first-order base, second-order corrected distribution, and first-order corrected particle constants. In the retained mixture example the particle coefficient increases by approximately 1.93. At equal NFE corrected Euler has only 3.3–4.7% smaller distribution error than base Heun; this does not imply lower wall-clock cost.

A correction is solver-specific. For $(aI + bJ)x$, Heun's leading distribution coefficient is proportional to $|a^3 - 3ab^2|$. The Euler choice $b = a$ worsens that coefficient relative to $b = 0$; the distinct choice $b = a/\sqrt{3}$ gives third-order distribution accuracy with second-order particle accuracy. Existing equal-NFE Heun/RK3 comparisons and all unfavorable particle results are retained. A fixed radial rotation mismatch restores first-order distribution error in the proven examples, although terminal cancellations can occur on other paths. Local necessity and a uniform-grid terminal-cancellation example are given in Appendix M.

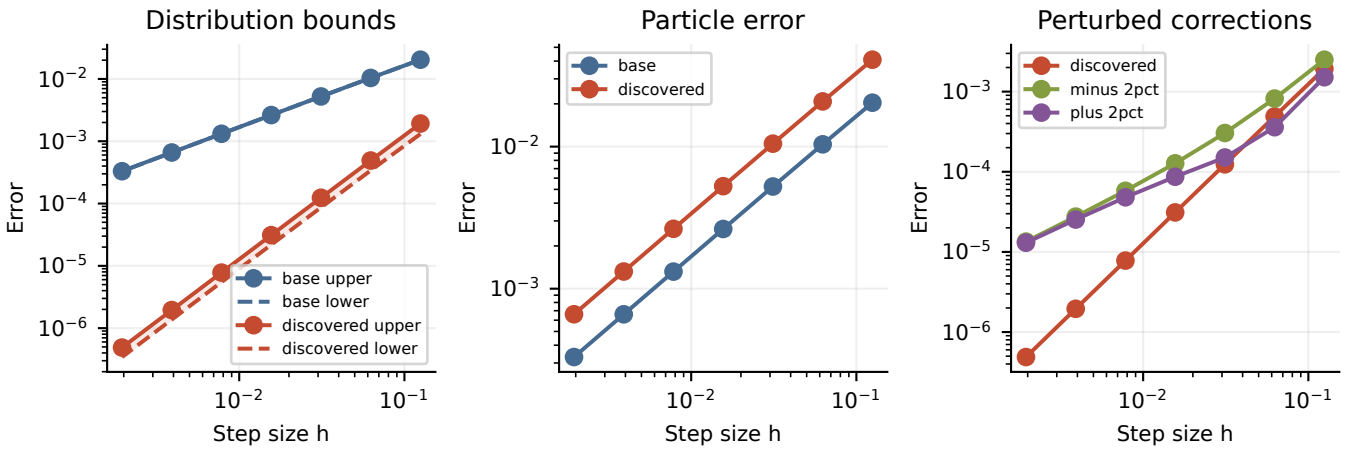


Fig. 1. Nonradial correction discovery followed by fixed-field refinement. Left: quadrature evaluations of lower and upper bounds on full W_2, with the region between them shaded. Center: common-start particle RMS. Right: fixed 2% perturbations of the discovered rotation restore an approximately first-order distribution-error trend. The corrected upper bound uses measure-preserving label rotation, not a same-start particle comparison.

VIII. Relation to Previous Work

Gauge freedom, weighted-gradient decompositions, and action minimization on a prescribed path are established in transport geometry and generative modeling [4–6, 8, 9]. The Gaussian rotation family is also known [10]. We do not claim the stream-function representation, the Hilbert projection theorem, or the convex chain rule as new. The constructive contribution is the reference-coordinate acceleration equation, its globally regular harmonic realization beyond affine-radial/product laws, and the exact correction certificate in that family. The strict quartic-observable counterexample supplies information not contained in scalar or covariance tests.

Hurault, Moreau, and Peyré [11, Theorem 1 and Eq. (4), v2] give weak Euler–Maruyama expansions for general smooth reverse diffusions, and specialize them to mean, covariance, and Fréchet errors. In their deterministic formula the propagated material acceleration controls the leading weak defect; our acceleration condition cancels it. The v2 analysis also develops Gaussian scale-mixture and parameter-selection expansions. Such weak and fitted-Gaussian discrepancies are distinct from full W_2 between non-Gaussian laws. Our finite-time coupling and projected-residual bounds control that full distance. The local coefficient theorem is a quantitative specialization of Wasserstein tangent geometry to Euler's defect, not a claim that tangent projection itself is new.

Tao and Choi [12] analyze numerical particle error through strain, vorticity, and acceleration. Their same-start pairing cannot remove the leading rotation error in our examples. Rectified-flow re-pairing and kinetic path design can change intermediate distributions [13–16]; our corrections preserve the stated path exactly. Bespoke solvers change the integrator [17], while exponential solvers exploit a chosen linear part [18, 19]. A matrix exponential is already exact for the constant linear Gaussian example.

Huang et al. [20] connect flow-matching and divergence losses to continuous-path error in total variation. That objective concerns velocity/path fidelity, rather than the projected material acceleration that controls Euler's integration defect here. Beyler and Bach [21] combine discretization, initialization, and score error in Wasserstein bounds for diffusion samplers, including Heun. Our target-path theorem uses a related stability principle while retaining the acceleration-projection cancellation. Convexity criteria for functionals on Wasserstein space [22] and higher invariant-measure orders for stochastic integrators [23, 24] are related background; neither is identified with our nonzero, distribution-preserving material acceleration on a changing path.

Recent work on Neural Wasserstein Splines [25] projects velocity and differentiates the moving projection to describe intrinsic Wasserstein acceleration. Our residual instead projects the material acceleration of the particular field being integrated. It is representation dependent, as the expanding Gaussian comparison demonstrates, and measures Euler's defect rather than an intrinsic acceleration of the marginal curve. The construction uses angular Fourier harmonics; it does not assert that its maps are spatially harmonic.

IX. Discussion and Limitations

The revised theory separates four regimes. Exact feasibility is demonstrated by the harmonic, Gaussian, and radial constructions. Certified approximation is described by upper residual bounds and the mixed $O(H^2) + O(\varepsilon H)$ guarantee. Proven infeasibility follows from convex-observable or covariance violations and can yield a strictly positive residual floor. For a general externally specified path, feasibility can remain unknown even when every tested moment condition passes.

The nonradial construction resolves the absence of an example beyond affine and product symmetry within a precise class; it does not establish a universal correction procedure. Its inverse-coordinate evaluation is additional computational work. A finite projection or small divergence penalty alone does not certify a small canonical residual. A correction selected separately as H changes yields a statement about a uniformly controlled family, rather than second-order accuracy of one fixed learned field. The learned-path theorem requires regularity,

moments, and a controlled mixed jerk term in addition to training loss. Global Lipschitz constants can make the bounds conservative.

The experiments support a mathematical mechanism and its certificates. They do not establish a practical speedup over Heun or RK3, scalable optimization for neural velocity fields, or improvements on image-generation benchmarks. Common-start particle errors remain important for inversion, editing, and paired transport. The new Gaussian-start variant, broader harmonic families, and efficient correction methods for externally given paths warrant further investigation.

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